

An Exact Four-State Theory of Feedback Control with Epistemic Resistance

A pedagogical Stage-I model for sensing, conditional mutual information, susceptibility, and information-response bounds

Working theory note

26 September 2026

Abstract

We develop a deliberately small discrete model for a population subject to feedback control. The population has two coarse variables: a binary social state X_t describing support for the controller target and a binary epistemic state K_t describing whether the population is in a high-information regime. A noisy sensor produces Y_t , a stochastic controller chooses an action U_t , and the next-day state $S_{t+1} = (X_{t+1}, K_{t+1})$ follows an exactly specified Markov kernel. Because the state space contains only four states, all finite-time distributions and information quantities are available without Monte Carlo simulation. The note derives, step by step, the sensing information $I(S_t; Y_t)$, the information actually used by the policy $I(S_t; U_t)$, the action-to-population conditional mutual information $I(U_t; X_{t+1} \mid X_t, K_t)$, the local susceptibility, and a decomposition of conditional action information into an average response term and an epistemic-modulation term. Two rigorous Pinsker-type inequalities then give bounded information-response efficiencies. The purpose is not yet to claim a literal thermodynamic efficiency, but to construct an exact reference system in the spirit of information-thermodynamic feedback bounds and stochastic thermodynamics without an energetic interpretation.

1 Motivation and scope

The long-term objective is to understand resource-constrained feedback control of interacting language-model agents whose decisions depend on both social information and distributed task evidence. Before constructing a mean-field theory for explicit fact spreading, it is useful to have a model small enough that every probability and information quantity can be written down exactly.

The present note is therefore a *Stage-I dummy theory*. It follows the same conceptual ordering as the feedback-information literature: define the stochastic dynamics first, identify what is measured, identify how the measurement changes the policy, and only then ask which information quantity can bound a useful response. Horowitz and Sandberg emphasize that different information measures correspond to different operational measurement scenarios, and hence different numerical “efficiencies” need not have identical physical meaning [1]. Recent work on stochastic thermodynamics of social imitation likewise shows that thermodynamic structure can be exported to social stochastic processes without assigning literal energies, provided the microscopic transition structure is specified carefully [2].

Accordingly, throughout this note we distinguish two claims:

1. **Exact information-response statements.** These follow from probability theory and information inequalities and are rigorous for the model below.
2. **Thermodynamic interpretation.** This requires an explicit reverse process or generalized local detailed balance construction and is postponed to a later stage.

All logarithms are natural unless stated otherwise, so information is measured in *nats*. This convention is important for the constants in Pinsker's inequality.

2 The four-state day/night model

2.1 Population state

At the end of day t the coarse population state is

$$S_t = (X_t, K_t), \quad (1)$$

with

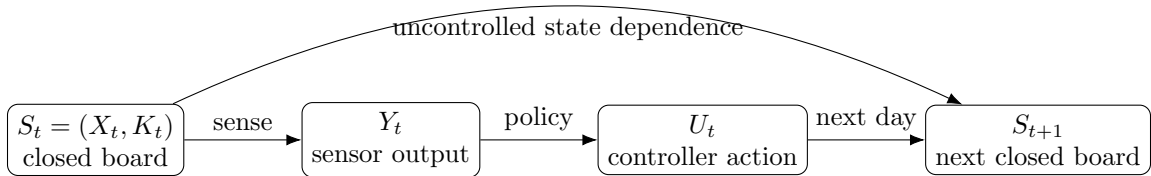
$$X_t \in \{0, 1\}, \quad K_t \in \{0, 1\}. \quad (2)$$

We interpret

- $X_t = 1$: the population supports the controller's target;
- $X_t = 0$: the population opposes the controller's target;
- $K_t = 1$: the population is in a high-epistemic-information state;
- $K_t = 0$: the population is in a low-information state.

For a false-targeting controller, $K = 1$ will be chosen to resist control.

The round clock is



Thus the controller senses after a day closes, chooses an action, and its intervention affects the following day.

2.2 Binary sensing channel

The controller does not see S_t exactly. It receives a binary observation $Y_t \in \{0, 1\}$ through a binary symmetric channel:

$$\mathbb{P}(Y_t = X_t \mid X_t) = 1 - \epsilon, \quad \mathbb{P}(Y_t \neq X_t \mid X_t) = \epsilon, \quad (3)$$

with $0 \leq \epsilon \leq 1/2$. The sensor reads the social state only, not the epistemic state:

$$\mathbb{P}(Y_t = y \mid X_t = x, K_t = k) = \mathbb{P}(Y_t = y \mid X_t = x). \quad (4)$$

This conditional independence will be the key to the sensing-information derivation.

2.3 Stochastic feedback policy

The controller maps the sensor output to a binary action $U_t \in \{0, 1\}$:

$$\mathbb{P}(U_t = 1 \mid Y_t = 0) = \pi_0, \quad \mathbb{P}(U_t = 1 \mid Y_t = 1) = \pi_1. \quad (5)$$

For a controller that intervenes mostly when support for its target is low, one expects $\pi_0 > \pi_1$.

After marginalizing the sensor, the action probability conditional on the true social state is

$$a_0 \equiv \mathbb{P}(U_t = 1 \mid X_t = 0) = (1 - \epsilon)\pi_0 + \epsilon\pi_1, \quad (6)$$

$$a_1 \equiv \mathbb{P}(U_t = 1 \mid X_t = 1) = \epsilon\pi_0 + (1 - \epsilon)\pi_1. \quad (7)$$

Because the sensor ignores K_t , U_t is conditionally independent of K_t given X_t :

$$U_t \perp K_t \mid X_t. \quad (8)$$

2.4 Controlled social dynamics

The next social state is Bernoulli with

$$p_{xku} \equiv \mathbb{P}(X_{t+1} = 1 \mid X_t = x, K_t = k, U_t = u) = \sigma[\theta + J(2x - 1) - gk + hu], \quad (9)$$

where $\sigma(z) = (1 + e^{-z})^{-1}$. The parameters are deliberately interpretable:

- J : persistence of the current social state;
- $g > 0$: epistemic resistance to a false controller;
- $h > 0$: actuation strength;
- θ : baseline bias.

No physical energy is being assigned to the logistic field in Eq. (9); it is simply a convenient stochastic response kernel.

2.5 Coarse epistemic dynamics

The epistemic state follows a two-state acquisition/forgetting process:

$$\mathbb{P}(K_{t+1} = 1 \mid K_t = 0) = \alpha, \quad \mathbb{P}(K_{t+1} = 0 \mid K_t = 1) = \delta. \quad (10)$$

Hence

$$\kappa^* \equiv \mathbb{P}(K = 1)_{\text{stat}} = \frac{\alpha}{\alpha + \delta}. \quad (11)$$

In the later mean-field theory, α and δ will be replaced by effective rates generated by fact transmission and persistence.

2.6 Exact four-by-four transition matrix

Order the four states as

$$(0, 0), \quad (0, 1), \quad (1, 0), \quad (1, 1). \quad (12)$$

The only potentially confusing notation in the matrix below is the shorthand A_{xk} and B_{xk} . We therefore derive those quantities before writing the matrix.

First average over the controller action

At a fixed current state $(X_t, K_t) = (x, k)$, the next social state still depends on whether the controller acts. Recall

$$p_{xku} = \mathbb{P}(X_{t+1} = 1 \mid X_t = x, K_t = k, U_t = u). \quad (13)$$

The controller acts with probability

$$a_x = \mathbb{P}(U_t = 1 \mid X_t = x). \quad (14)$$

Therefore, after averaging over the stochastic controller action,

$$\bar{p}_{xk} \equiv \mathbb{P}(X_{t+1} = 1 \mid X_t = x, K_t = k) \quad (15)$$

$$= (1 - a_x)p_{xk0} + a_x p_{xk1}. \quad (16)$$

This is simply a two-component mixture: with probability $1 - a_x$ the controller is silent and the transition probability is p_{xk0} ; with probability a_x it acts and the transition probability is p_{xk1} .

We now introduce the abbreviations

$$\boxed{B_{xk} \equiv \bar{p}_{xk}} \quad \text{and} \quad \boxed{A_{xk} \equiv 1 - \bar{p}_{xk}}. \quad (17)$$

Thus they have no additional dynamical meaning:

$$B_{xk} = \mathbb{P}(X_{t+1} = 1 \mid X_t = x, K_t = k), \quad (18)$$

$$A_{xk} = \mathbb{P}(X_{t+1} = 0 \mid X_t = x, K_t = k). \quad (19)$$

The letters A and B are used only to keep the matrix compact.

For example, at current state $(x, k) = (0, 1)$,

$$B_{01} = (1 - a_0)p_{010} + a_0 p_{011}, \quad (20)$$

where p_{010} is the probability of target support tomorrow if the controller is silent, and p_{011} is the corresponding probability if it intervenes. Then

$$A_{01} = 1 - B_{01}. \quad (21)$$

Combine the social and epistemic transitions

The Stage-I model assumes that, conditional on the current state and after averaging over U_t , the next social and epistemic coordinates factorize:

$$\mathbb{P}(X_{t+1} = x', K_{t+1} = k' \mid X_t = x, K_t = k) = \mathbb{P}(X_{t+1} = x' \mid x, k) \mathbb{P}(K_{t+1} = k' \mid k). \quad (22)$$

For instance, starting from $(0, 0)$,

$$\mathbb{P}((0, 0) \rightarrow (0, 0)) = A_{00}(1 - \alpha), \quad (23)$$

$$\mathbb{P}((0, 0) \rightarrow (0, 1)) = A_{00}\alpha, \quad (24)$$

$$\mathbb{P}((0, 0) \rightarrow (1, 0)) = B_{00}(1 - \alpha), \quad (25)$$

$$\mathbb{P}((0, 0) \rightarrow (1, 1)) = B_{00}\alpha. \quad (26)$$

The first factor in each product chooses the next social state; the second factor chooses the next epistemic state.

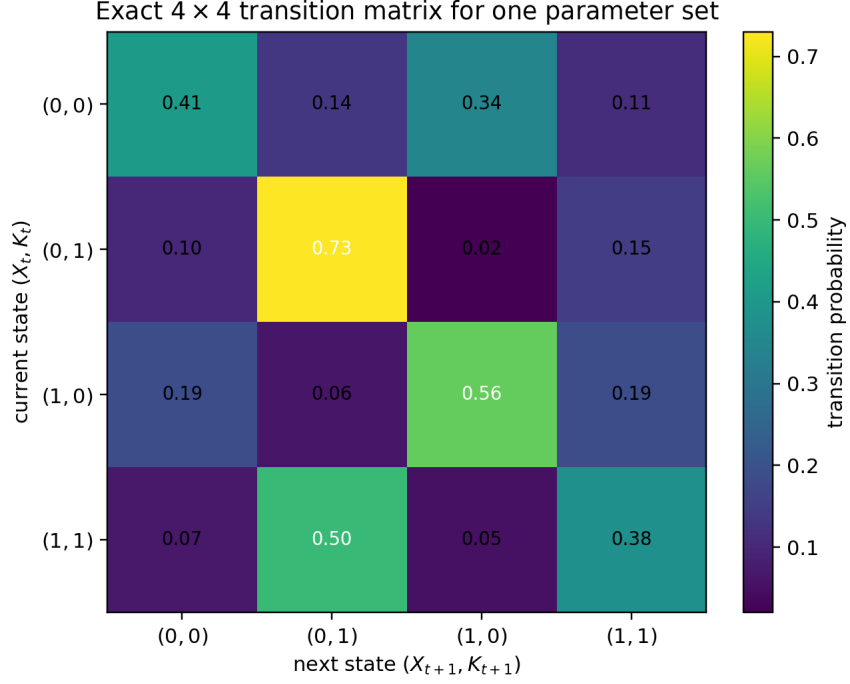


Figure 1: Heatmap of the exact 4×4 transition matrix for one concrete parameter set. Row i corresponds to a current state (X_t, K_t) and column j to a next state (X_{t+1}, K_{t+1}) . The entries are ordinary probabilities, so each row sums to one.

With this notation the exact transition matrix is

$$M = \begin{pmatrix} A_{00}(1 - \alpha) & A_{00}\alpha & B_{00}(1 - \alpha) & B_{00}\alpha \\ A_{01}\delta & A_{01}(1 - \delta) & B_{01}\delta & B_{01}(1 - \delta) \\ A_{10}(1 - \alpha) & A_{10}\alpha & B_{10}(1 - \alpha) & B_{10}\alpha \\ A_{11}\delta & A_{11}(1 - \delta) & B_{11}\delta & B_{11}(1 - \delta) \end{pmatrix}. \quad (27)$$

If one prefers to avoid the A/B shorthand completely, the first row is equivalently

$$[(1 - \bar{p}_{00})(1 - \alpha), (1 - \bar{p}_{00})\alpha, \bar{p}_{00}(1 - \alpha), \bar{p}_{00}\alpha], \quad (28)$$

and the remaining rows follow in exactly the same way.

Finally, if

$$\boldsymbol{\mu}_t = (\mathbb{P}_t(0, 0), \mathbb{P}_t(0, 1), \mathbb{P}_t(1, 0), \mathbb{P}_t(1, 1)), \quad (29)$$

then

$$\boxed{\boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t M.} \quad (30)$$

Thus all finite-time observables are obtained exactly by matrix multiplication.

Figures 1–4 make this concrete. Figure 1 shows the matrix itself for one representative parameter set. Each row is an outgoing probability distribution from the corresponding current state, so every row sums to one. Figure 2 presents the same information as a state-flow diagram: the matrix is nothing more than a compact way to list the four outgoing arrow distributions at once.

A second point that is easy to miss in the algebra is that the matrix acts on an entire *distribution*, not on one state label. If the current distribution is $\boldsymbol{\mu}_t$, then multiplying by M yields the next distribution $\boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t M$. This means that probability mass is redistributed simultaneously across all four states. Figure 3 shows this explicitly by comparing $\boldsymbol{\mu}_t$, $\boldsymbol{\mu}_{t+1}$, and $\boldsymbol{\mu}_{t+2}$ for one initial condition.

State-flow view: one row of the matrix is an outgoing probability distribution

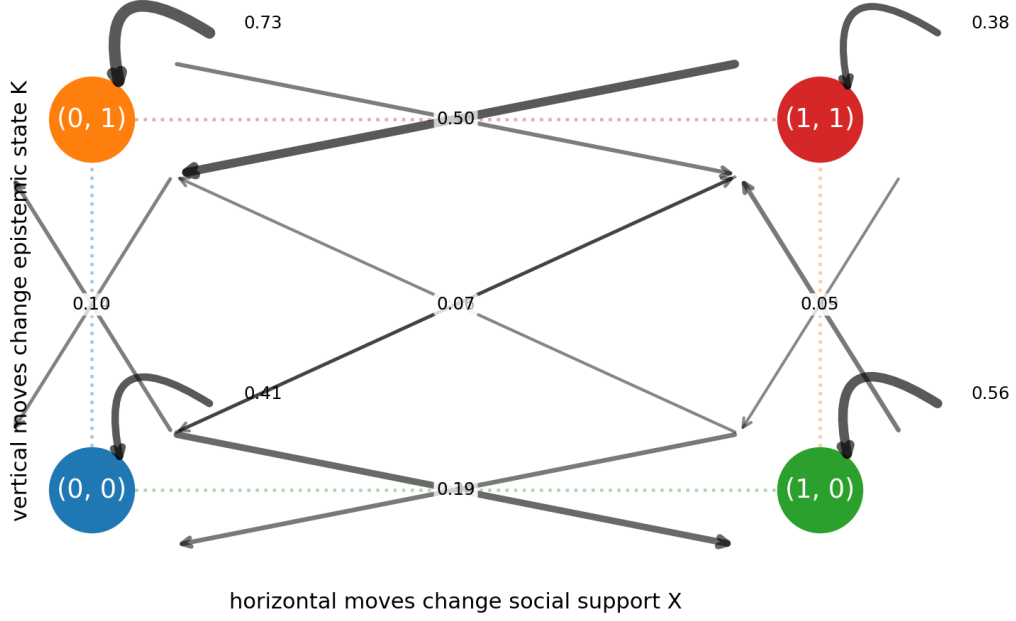


Figure 2: State-flow representation of the same matrix. Each node is one of the four possible states. The arrows leaving a node show exactly the corresponding row of the transition matrix. Horizontal movement changes the social variable X , while vertical movement changes the epistemic variable K .

Finally, one should think of the matrix as a compact fingerprint of the parameter regime. Changing a parameter such as the epistemic-resistance strength g changes the whole matrix, not just a single entry. Figure 4 compares two matrices with weak and strong epistemic resistance. When g is larger, transitions into target-supporting states become more suppressed from high-information rows, which is exactly the kind of qualitative behavior we later describe as epistemic resistance to control.

3 Sensing mutual information, derived step by step

This section derives the formula that is easy to quote but easy to lose intuition for:

$$I_{\text{sens}}(t) = I(S_t; Y_t). \quad (31)$$

3.1 Step 1: start from the definition

For arbitrary discrete variables A and B ,

$$I(A; B) = \sum_{a,b} p(a,b) \ln \frac{p(a,b)}{p(a)p(b)} = \sum_{a,b} p(a,b) \ln \frac{p(b|a)}{p(b)}. \quad (32)$$

Rearranging the logarithm gives the entropy identity

$$I(A; B) = H(B) - H(B | A). \quad (33)$$

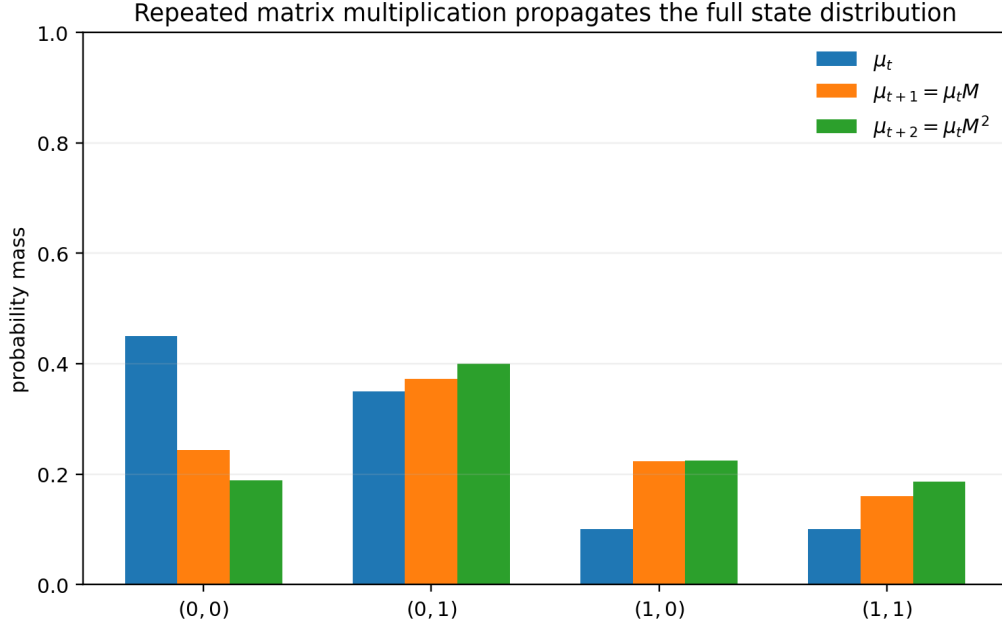


Figure 3: Example of exact propagation of the four-state probability vector. Starting from an initial distribution μ_t , one multiplication by M gives μ_{t+1} and a second multiplication gives μ_{t+2} . This is the operational meaning of the Markov update rule $\mu_{t+1} = \mu_t M$.

For the sensor,

$$I(S_t; Y_t) = H(Y_t) - H(Y_t | S_t). \quad (34)$$

This already gives the interpretation: sensing information is the uncertainty in the sensor output minus the uncertainty that remains once the true state is known.

3.2 Step 2: why the epistemic variable disappears from the sensor term

Because $S_t = (X_t, K_t)$,

$$I(S_t; Y_t) = I(X_t, K_t; Y_t). \quad (35)$$

The chain rule for mutual information gives

$$I(X_t, K_t; Y_t) = I(X_t; Y_t) + I(K_t; Y_t | X_t). \quad (36)$$

But Eq. (4) says that once X_t is known, the sensor output carries no further information about K_t . Therefore

$$I(K_t; Y_t | X_t) = 0, \quad (37)$$

and hence

$$\boxed{I(S_t; Y_t) = I(X_t; Y_t)}. \quad (38)$$

This equality is not a generic property of epistemic feedback systems. It follows specifically from our modeling choice that the controller senses only the social state.

3.3 Step 3: compute the sensor-output distribution

Let

$$q_t \equiv \mathbb{P}(X_t = 1). \quad (39)$$

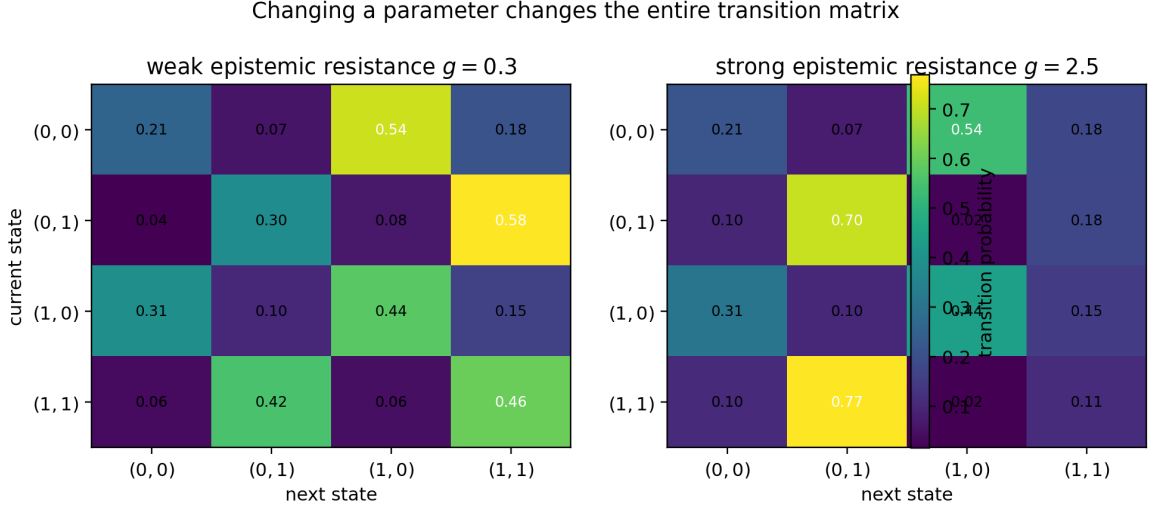


Figure 4: Two concrete matrices for two values of the epistemic-resistance parameter g . The matrix therefore serves as a complete fingerprint of the one-step dynamics: changing model parameters changes the whole pattern of one-step transitions.

The probability that the sensor reports $Y_t = 1$ is obtained by the law of total probability:

$$r_t \equiv \mathbb{P}(Y_t = 1) \quad (40)$$

$$= \mathbb{P}(Y_t = 1 \mid X_t = 1)\mathbb{P}(X_t = 1) + \mathbb{P}(Y_t = 1 \mid X_t = 0)\mathbb{P}(X_t = 0) \quad (41)$$

$$= (1 - \epsilon)q_t + \epsilon(1 - q_t) \quad (42)$$

$$= \boxed{\epsilon + (1 - 2\epsilon)q_t}. \quad (43)$$

Thus Y_t is Bernoulli with parameter r_t .

3.4 Step 4: compute the two entropies

Define the binary entropy in nats

$$h(p) = -p \ln p - (1 - p) \ln(1 - p). \quad (44)$$

Since $Y_t \sim \text{Bernoulli}(r_t)$,

$$H(Y_t) = h(r_t). \quad (45)$$

Conditional on either value of X_t , the sensor flips the correct value with probability ϵ . Therefore

$$H(Y_t \mid X_t = x) = h(\epsilon) \quad \text{for both } x = 0, 1. \quad (46)$$

Averaging over X_t changes nothing:

$$H(Y_t \mid S_t) = H(Y_t \mid X_t) = h(\epsilon). \quad (47)$$

3.5 Step 5: closed form

Substituting into Eq. (34),

$$\boxed{I_{\text{sens}}(t) = I(S_t; Y_t) = h(\epsilon + (1 - 2\epsilon)q_t) - h(\epsilon)}. \quad (48)$$

This is exact for any distribution over the four states; only the marginal $q_t = P(X_t = 1)$ is needed.

Two checks are useful:

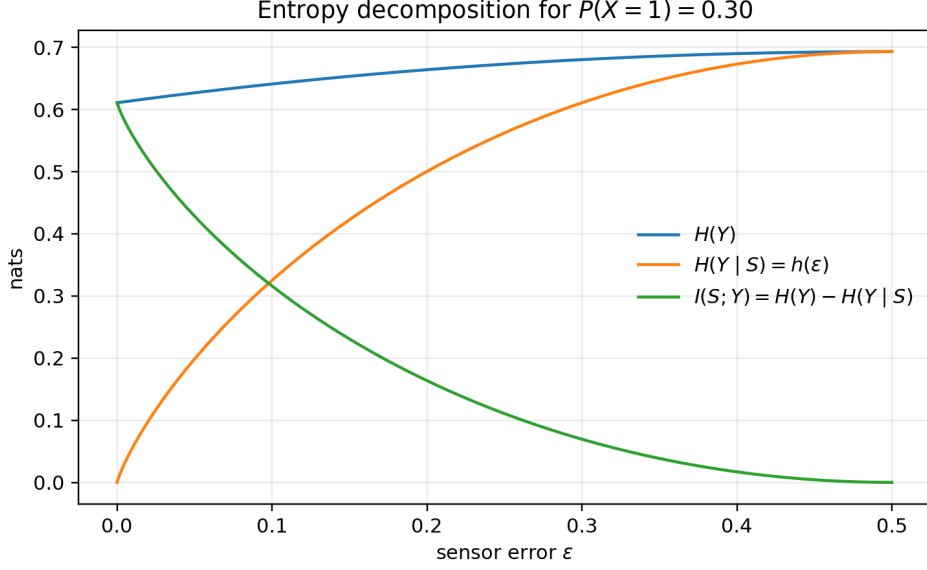


Figure 5: The sensing information is literally the vertical difference between the output entropy $H(Y)$ and the residual sensor noise $H(Y | S)$. The example uses $P(X = 1) = 0.30$.

- If $\epsilon = 1/2$, then Y_t is independent random noise and $I_{\text{sens}} = 0$.
- If $\epsilon = 0$, then $Y_t = X_t$ exactly and $I_{\text{sens}} = H(X_t) = h(q_t)$.

4 How much of the sensed information is actually used?

Sensing does not imply that the controller uses all available information. The effective state-dependent action probabilities are a_0, a_1 from Eq. (7). With $q_t = P(X_t = 1)$, the marginal intervention probability is

$$\bar{a}_t = (1 - q_t)a_0 + q_t a_1. \quad (49)$$

Since U_t is Bernoulli,

$$H(U_t) = h(\bar{a}_t). \quad (50)$$

Conditioned on the social state,

$$H(U_t | X_t) = (1 - q_t)h(a_0) + q_t h(a_1). \quad (51)$$

Because $U_t \perp K_t | X_t$,

$$I(S_t; U_t) = I(X_t; U_t) = h(\bar{a}_t) - (1 - q_t)h(a_0) - q_t h(a_1). \quad (52)$$

The Markov chain $S_t \rightarrow Y_t \rightarrow U_t$ implies by the data-processing inequality

$$I(S_t; U_t) \leq I(S_t; Y_t). \quad (53)$$

Thus $I(S; Y)$ is acquired state information, whereas $I(S; U)$ is the portion that survives the policy map and becomes state-dependent action.

Figure 7 makes this distinction concrete. Parameterize the policy by a contrast $d \in [0, 1]$,

$$\pi_0 = \frac{1 + d}{2}, \quad \pi_1 = \frac{1 - d}{2}. \quad (54)$$

At $d = 0$ the controller ignores the sensor and acts with probability $1/2$, so $I(S; U) = 0$. At $d = 1$ the action is a deterministic relabeling of the sensor output, so no additional information is lost and $I(S; U) = I(S; Y)$.

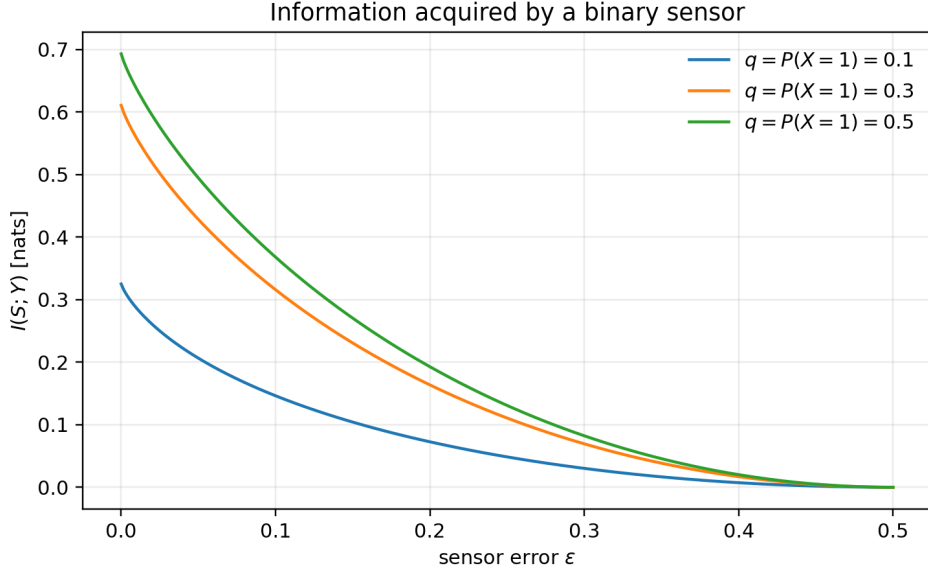


Figure 6: Exact sensing mutual information from Eq. (48). A balanced social state carries more potential information than a nearly deterministic social state, and all sensing information vanishes at $\epsilon = 1/2$.

5 Action-to-population conditional mutual information

The sensing information is about the direction $S_t \rightarrow Y_t$. The quantity used in the current control analysis asks the opposite operational question: once the current state is fixed, how much does the controller action tell us about the next population state?

We define

$$T_{\pi}^{X|K}(t) \equiv I(U_t; X_{t+1} \mid X_t, K_t) = I(U_t; X_{t+1} \mid S_t). \quad (55)$$

5.1 Exact local expression

Fix a state $s = (x, k)$. Write

$$a \equiv a_x, \quad p_0 \equiv p_{xk0}, \quad p_1 \equiv p_{xk1}. \quad (56)$$

Marginalizing over the controller action gives

$$\bar{p} = (1 - a)p_0 + ap_1 = \mathbb{P}(X_{t+1} = 1 \mid S_t = s). \quad (57)$$

The conditional mutual information at that state is

$$T_s \equiv I(U_t; X_{t+1} \mid S_t = s) \quad (58)$$

$$= H(X_{t+1} \mid S_t = s) - H(X_{t+1} \mid U_t, S_t = s) \quad (59)$$

$$= \boxed{h(\bar{p}) - (1 - a)h(p_0) - ah(p_1)}. \quad (60)$$

The state-averaged quantity is therefore

$$\boxed{T_{\pi}^{X|K}(t) = \sum_{x,k} \mu_t(x, k) T_{xk}}. \quad (61)$$

No estimator is needed in this four-state model. Equation (61) is the exact population value against which a finite-sample estimator can later be tested.

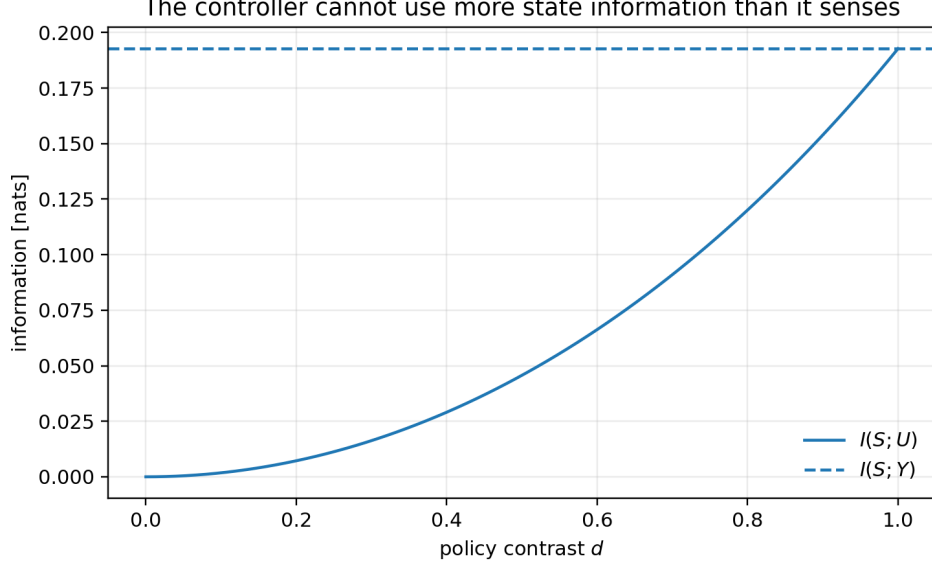


Figure 7: Information available to the controller versus information expressed in its action for $\epsilon = 0.2$ and $P(X = 1) = 0.5$. The policy cannot create state information that was absent from the sensor.

5.2 Why condition on the epistemic state?

One may instead calculate

$$T_{\pi}^X(t) = I(U_t; X_{t+1} | X_t), \quad (62)$$

which mixes high- and low-information populations having the same social state. The difference between the two quantities follows exactly from the chain rule. On one hand,

$$I(U; X', K | X) = I(U; K | X) + I(U; X' | X, K). \quad (63)$$

Equation (8) sets the first term to zero, so this equals $I(U; X' | X, K)$. On the other hand,

$$I(U; X', K | X) = I(U; X' | X) + I(U; K | X, X'). \quad (64)$$

Equating the two expansions gives

$$\boxed{I(U; X' | X, K) = I(U; X' | X) + I(U; K | X, X') \geq I(U; X' | X)}. \quad (65)$$

The extra term is not a nuisance: it quantifies precisely how much controller-action information is hidden when populations with different epistemic states are pooled together. This is the theoretical reason to condition empirical CMI on a coarse epistemic coordinate.

6 Susceptibility: the coherent response to intervention

At fixed state $s = (x, k)$ define the local susceptibility

$$\chi_{xk} \equiv \mathbb{E}[X_{t+1} | x, k, U_t = 1] - \mathbb{E}[X_{t+1} | x, k, U_t = 0] \quad (66)$$

$$= \boxed{p_{xk1} - p_{xk0}}. \quad (67)$$

For the logistic kernel, let

$$z_{xk} = \theta + J(2x - 1) - gk. \quad (68)$$

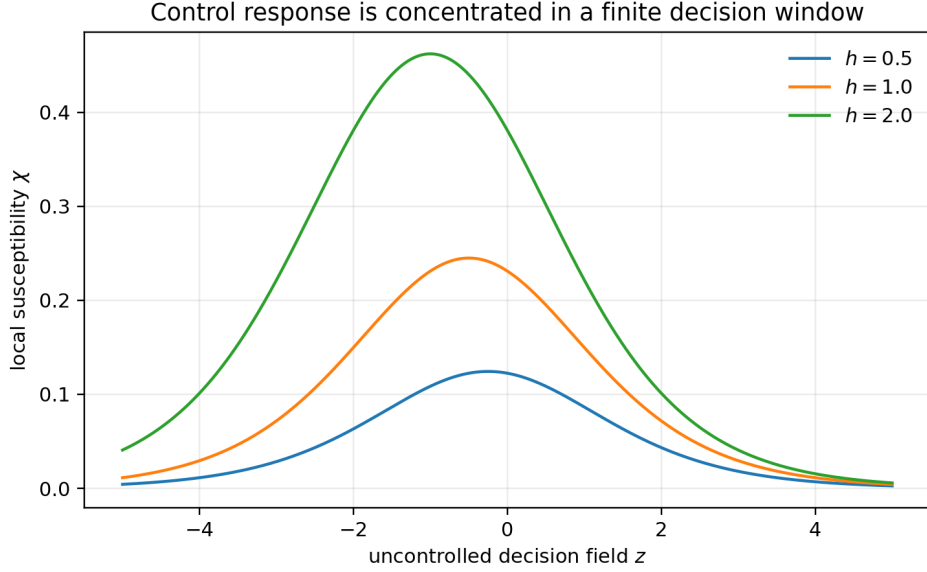


Figure 8: Exact local susceptibility for the logistic controlled transition. Strong response occurs only in a finite window of the uncontrolled decision field. This is why large actuation or strong epistemic bias can reduce useful response by pushing the system into saturation.

Then

$$\boxed{\chi_{xk} = \sigma(z_{xk} + h) - \sigma(z_{xk})}. \quad (69)$$

For weak actuation h ,

$$\chi_{xk} = h\sigma(z_{xk})[1 - \sigma(z_{xk})] + \mathcal{O}(h^2). \quad (70)$$

Hence response is small when the next decision is already almost deterministic and largest in an intermediate decision region. For finite h , the maximum is centered near $z = -h/2$.

7 A rigorous information-response bound

The conditional mutual information in Eq. (60) measures the full distributional distinction between action-conditioned next states. Susceptibility measures only their coherent difference in mean target support. Pinsker’s inequality relates the two.

7.1 Step 1: write local mutual information as a KL mixture

At a fixed state s , let P_0 and P_1 denote the distributions of X_{t+1} under $U = 0$ and $U = 1$, and let

$$\bar{P} = (1 - a)P_0 + aP_1. \quad (71)$$

Then

$$I(U; X' | s) = (1 - a)D_{\text{KL}}(P_0 \| \bar{P}) + aD_{\text{KL}}(P_1 \| \bar{P}). \quad (72)$$

7.2 Step 2: apply Pinsker’s inequality

In nats,

$$D_{\text{KL}}(P \| Q) \geq 2\text{TV}(P, Q)^2. \quad (73)$$

Because $\bar{P} = (1 - a)P_0 + aP_1$,

$$\text{TV}(P_0, \bar{P}) = a \text{TV}(P_0, P_1), \quad (74)$$

$$\text{TV}(P_1, \bar{P}) = (1 - a) \text{TV}(P_0, P_1). \quad (75)$$

Substituting these into Eq. (72) yields

$$I(U; X' | s) \geq 2a(1 - a)\text{TV}(P_0, P_1)^2. \quad (76)$$

For a binary next state, total variation is exactly the absolute difference of the Bernoulli probabilities:

$$\text{TV}(P_0, P_1) = |p_1 - p_0| = |\chi_s|. \quad (77)$$

Therefore

$$\boxed{I(U; X' | S = s) \geq 2a_s(1 - a_s)\chi_s^2.} \quad (78)$$

7.3 Step 3: average over states

Averaging Eq. (78) over S_t gives

$$\boxed{T_\pi^{X|K}(t) \geq 2 \sum_s \mu_t(s) a_s (1 - a_s) \chi_s^2.} \quad (79)$$

This motivates the bounded information-response efficiency

$$\boxed{\eta_{\text{IR}}(t) \equiv \frac{2 \sum_s \mu_t(s) a_s (1 - a_s) \chi_s^2}{I(U_t; X_{t+1} | S_t)} \leq 1.} \quad (80)$$

It asks: *what fraction of the action-dependent distributional information is expressed as a coherent mean shift in the chosen population observable?*

Bits versus nats. Empirical analyses often report MI in bits. Since $I_{\text{nats}} = (\ln 2) I_{\text{bits}}$, Eq. (78) becomes

$$I_{\text{bits}}(U; X' | s) \geq \frac{2}{\ln 2} a(1 - a) \chi_s^2. \quad (81)$$

The conversion factor must be included before comparing a bit-valued empirical CMI with the Pinsker numerator.

8 How epistemic state modulates controllability

The previous section bound the controller's mean effect at each full state $s = (x, k)$. A separate and scientifically central question is whether *the same controller action has different consequences depending on what the population knows*. This is the point of conditioning on the epistemic state in

$$I(U_t; X_{t+1} | X_t, K_t). \quad (82)$$

Here we make that statement quantitative.

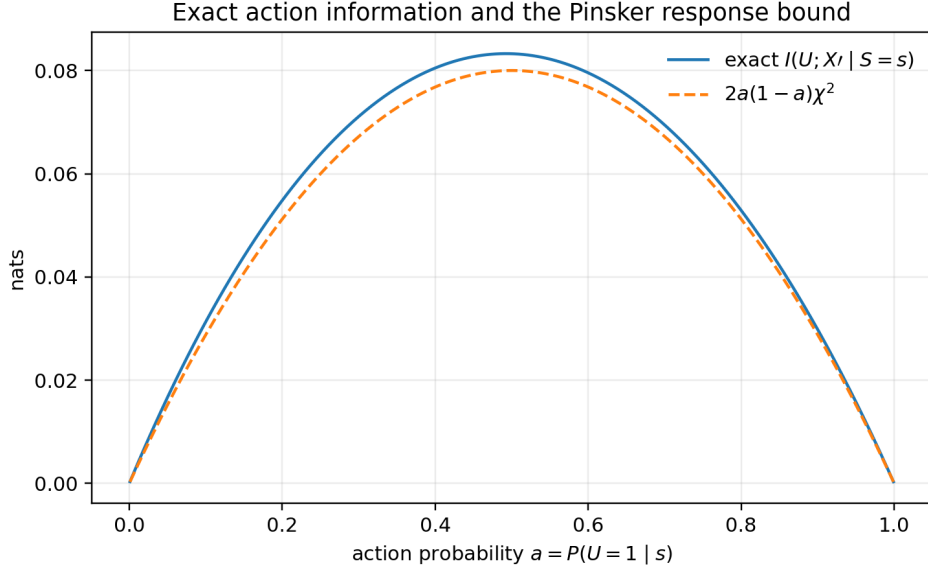


Figure 9: Example of Eq. (78) with $p_0 = 0.25$ and $p_1 = 0.65$. The exact conditional mutual information lies above the Pinsker response term for every action probability. Both vanish when the policy almost never or almost always takes one action, illustrating the importance of action support.

8.1 Step 1: local susceptibilities at fixed social state

Fix the current social state $X_t = x$. There are then two relevant local susceptibilities,

$$\chi_{x0} = \mathbb{E}[X_{t+1} | X_t = x, K_t = 0, U_t = 1] - \mathbb{E}[X_{t+1} | X_t = x, K_t = 0, U_t = 0], \quad (83)$$

and

$$\chi_{x1} = \mathbb{E}[X_{t+1} | X_t = x, K_t = 1, U_t = 1] - \mathbb{E}[X_{t+1} | X_t = x, K_t = 1, U_t = 0]. \quad (84)$$

If these are different, the controller’s ability to move the population depends on epistemic state. Figure 10 shows the basic phenomenon in the logistic model: as the epistemic-resistance parameter g increases, the high-information state $K = 1$ becomes increasingly resistant, and the response gap

$$\Delta\chi_x \equiv \chi_{x1} - \chi_{x0} \quad (85)$$

grows in magnitude.

8.2 Step 2: average over the epistemic state at fixed social state

Let

$$r_x \equiv \mathbb{P}(K_t = 1 | X_t = x) \quad (86)$$

be the probability that a population with social state x is in the high-information regime. The epistemic-state-averaged susceptibility at that social state is

$$\bar{\chi}_x \equiv \mathbb{E}[\chi_{xK} | X_t = x] = (1 - r_x)\chi_{x0} + r_x\chi_{x1}. \quad (87)$$

Now use the elementary variance identity

$$\mathbb{E}[\chi_{xK}^2 | X_t = x] = \bar{\chi}_x^2 + \text{Var}(\chi_{xK} | X_t = x). \quad (88)$$

Because K is binary,

$$\text{Var}(\chi_{xK} | X_t = x) = r_x(1 - r_x)(\chi_{x1} - \chi_{x0})^2. \quad (89)$$

This simple formula is already informative: epistemic modulation vanishes if either epistemic state never occurs ($r_x = 0$ or 1) or if the controller affects both epistemic states equally ($\chi_{x1} = \chi_{x0}$).

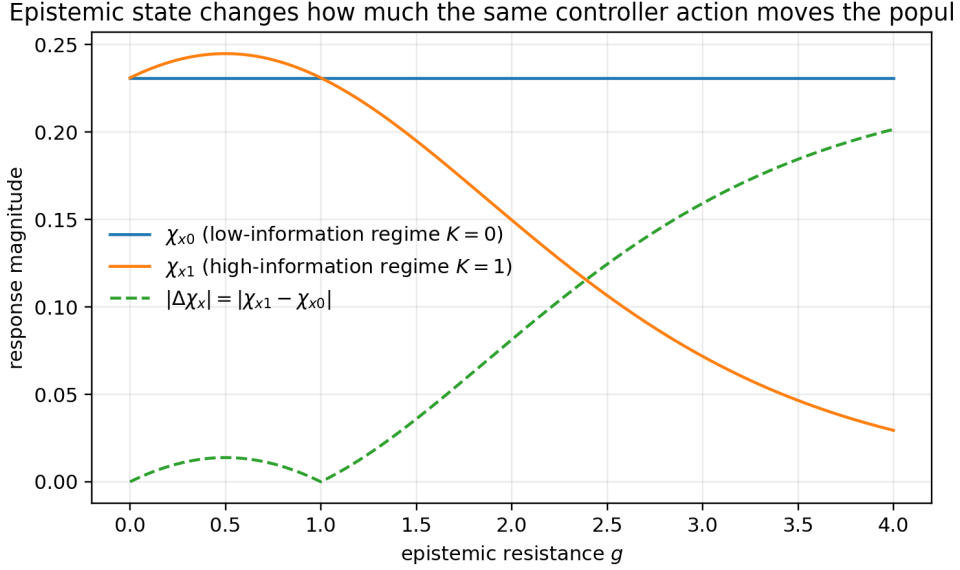


Figure 10: Epistemic modulation of local susceptibility for a fixed social state $x = 0$. The low-information and high-information populations respond differently to the same intervention. The dashed curve shows the modulation gap $|\Delta\chi_x| = |\chi_{x1} - \chi_{x0}|$.

8.3 Step 3: decompose the Pinsker lower bound

Equation (78) gives, at each full state $s = (x, k)$,

$$I(U; X' | X = x, K = k) \geq 2a_x(1 - a_x)\chi_{xk}^2. \quad (90)$$

Average this inequality first over K conditional on $X = x$:

$$\sum_k \mathbb{P}(k | x) I(U; X' | x, k) \geq 2a_x(1 - a_x) \sum_k \mathbb{P}(k | x) \chi_{xk}^2. \quad (91)$$

Now substitute Eq. (88):

$$\sum_k \mathbb{P}(k | x) I(U; X' | x, k) \geq 2a_x(1 - a_x) [\bar{\chi}_x^2 + \text{Var}(\chi_{xK} | x)]. \quad (92)$$

Finally average over the current social state X_t to obtain

$$I(U_t; X_{t+1} | X_t, K_t) \geq 2 \sum_x \mathbb{P}(X_t = x) a_x(1 - a_x) [\bar{\chi}_x^2 + \text{Var}(\chi_{xK} | X_t = x)]. \quad (93)$$

Using Eq. (89), this becomes

$$I(U_t; X_{t+1} | X_t, K_t) \geq 2 \sum_x \mathbb{P}(X_t = x) a_x(1 - a_x) [\bar{\chi}_x^2 + r_x(1 - r_x)(\chi_{x1} - \chi_{x0})^2]. \quad (94)$$

This is the missing bound we wanted: the conditional controller information must be large enough not only to support an average control response $\bar{\chi}_x$, but also to support the *epistemic dependence of that response*.

The two pieces have different meanings:

$$\text{mean-response piece} = 2 \sum_x \mathbb{P}(x) a_x(1 - a_x) \bar{\chi}_x^2, \quad (95)$$

$$\text{epistemic-modulation piece} = 2 \sum_x \mathbb{P}(x) a_x(1 - a_x) \text{Var}(\chi_{xK} | x). \quad (96)$$

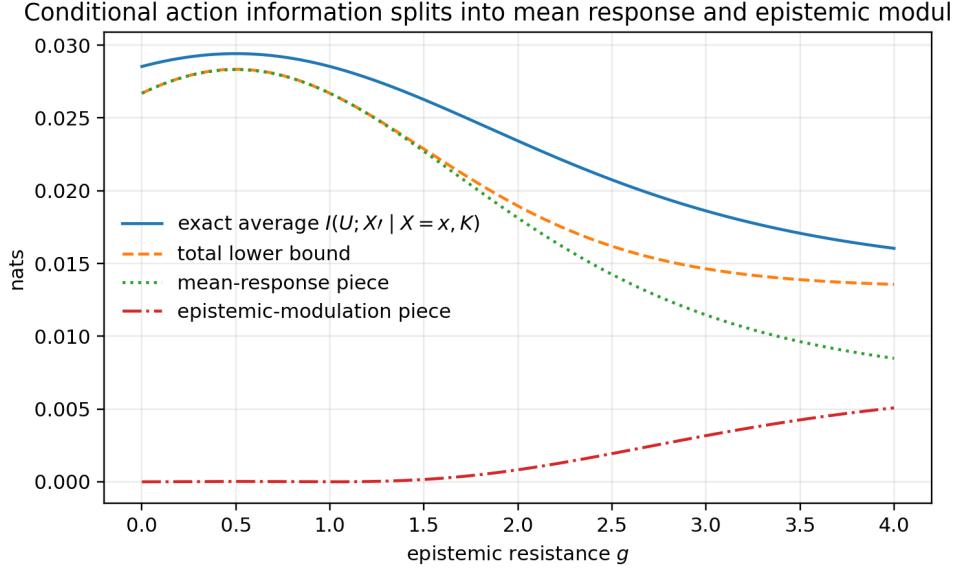


Figure 11: At a fixed social state x , the exact conditional action information (solid curve) exceeds the lower bound obtained by splitting the response into a mean-response contribution and an epistemic-modulation contribution. As epistemic resistance g increases, the controller affects low-information and high-information populations differently, and the modulation term grows.

The second term is the explicit measure of how strongly controllability varies across epistemic states.

8.4 Step 4: an epistemic modulation efficiency

Equation (93) suggests a bounded normalized quantity,

$$\eta_{\text{epi}}(t) \equiv \frac{2 \sum_x \mathbb{P}(X_t = x) a_x (1 - a_x) \text{Var}(\chi_{xK} | X_t = x)}{I(U_t; X_{t+1} | X_t, K_t)} \leq 1. \quad (97)$$

We do not give Eq. (97) a thermodynamic interpretation at this stage. Operationally it answers a clean question:

How much of the conditional controller information can be attributed specifically to the fact that the same action has different effects in different epistemic regimes?

This makes precise the informal idea of *epistemic resistance*: if high-information populations are harder to move than low-information populations, then the variability of χ_{xK} across K consumes part of the conditional action information budget.

8.5 Remark: state before or after the day?

In the present four-state model we conditioned on K_t , the epistemic state at the beginning of the round. In the full day/night dashboard system, information also spreads during the day before the controller acts. A more faithful later theory may therefore condition on a post-day epistemic variable, say $\tilde{K}_{t+1}$, and study

$$I(U_t; X_{t+1} | X_t, \tilde{K}_{t+1}). \quad (98)$$

The logic of the decomposition would be the same. We start with K_t here only because it keeps the finite-state model exact and transparent.

9 A second bound: the behavioral value of sensing

The previous bound starts from $I(U; X' | S)$ and asks how much action information appears as population response. A distinct question is: *how much advantage can feedback gain over an equally active open-loop controller simply because it knows the state?*

9.1 Matched open-loop reference

At a given round, define the effective feedback policy

$$\pi(u | s) = \mathbb{P}(U_t = u | S_t = s) \quad (99)$$

and its marginal action frequency

$$\bar{\pi}(u) = \sum_s \mu_t(s) \pi(u | s). \quad (100)$$

Construct a matched open-loop controller that uses the same marginal action frequency but chooses U independently of S .

The feedback and matched-open-loop state-action distributions are

$$P(s, u) = \mu_t(s) \pi(u | s), \quad (101)$$

$$Q(s, u) = \mu_t(s) \bar{\pi}(u). \quad (102)$$

Their relative entropy is exactly

$$D_{\text{KL}}(P \| Q) = \sum_{s, u} \mu_t(s) \pi(u | s) \ln \frac{\pi(u | s)}{\bar{\pi}(u)} \quad (103)$$

$$= \boxed{I(S_t; U_t)}. \quad (104)$$

Thus $I(S; U)$ is precisely the KL separation between state-dependent feedback actions and an equally active state-independent controller.

9.2 Propagate both controllers through the same dynamics

Let $W(s' | s, u)$ denote the next-state channel. Applying the same stochastic channel to P and Q cannot increase KL divergence, so

$$D_{\text{KL}}(P_{S'} \| Q_{S'}) \leq I(S_t; U_t). \quad (105)$$

For any bounded observable $f(S') \in [0, 1]$, the difference in expectation is bounded by total variation, and Pinsker gives

$$|\mathbb{E}_P[f] - \mathbb{E}_Q[f]| \leq \text{TV}(P_{S'}, Q_{S'}) \leq \sqrt{\frac{1}{2} D_{\text{KL}}(P_{S'} \| Q_{S'})}. \quad (106)$$

Taking $f(S') = X'$ and defining

$$\Delta_{\text{fb}} \equiv \mathbb{E}_{\text{feedback}}[X_{t+1}] - \mathbb{E}_{\text{matched open}}[X_{t+1}], \quad (107)$$

we obtain

$$\boxed{2\Delta_{\text{fb}}^2 \leq I(S_t; U_t) \leq I(S_t; Y_t)}. \quad (108)$$

The second inequality is Eq. (53).

Equation (108) is particularly useful conceptually. It says that the extra one-step behavioral effect obtained from *state-dependent* intervention, relative to an equally active open-loop controller, cannot exceed the information resource that the feedback loop actually uses, and that used information cannot exceed what was sensed.

Two bounded ratios follow naturally:

$$\eta_{\text{used}} = \frac{2\Delta_{\text{fb}}^2}{I(S;U)} \leq 1, \quad \eta_{\text{sens}} = \frac{2\Delta_{\text{fb}}^2}{I(S;Y)} \leq 1. \quad (109)$$

These are information-response efficiencies, not energetic efficiencies.

10 Exact numerical example

To make the formulas concrete, consider the parameter set in Table 1. The controller target is interpreted as false, so $g > 0$ represents epistemic resistance.

Table 1: Reference parameter set used in the figures below.

Parameter	Value	Meaning
ϵ	0.20	sensor error
π_0, π_1	0.85, 0.15	feedback policy
θ	-0.20	baseline social bias
J	1.00	social persistence
g	1.50	epistemic resistance
h	1.40	actuation strength
α	0.25	information acquisition
δ	0.12	information loss
μ_0	(0.45, 0.35, 0.10, 0.10)	initial four-state distribution

The effective action probabilities are

$$a_0 = 0.71, \quad a_1 = 0.29. \quad (110)$$

Initially $q_0 = P(X_0 = 1) = 0.20$ and $\kappa_0 = P(K_0 = 1) = 0.45$. Exact matrix propagation gives the trajectories in Fig. 12. The epistemic state relaxes to the analytically known value

$$\kappa^* = \frac{0.25}{0.25 + 0.12} \approx 0.676, \quad (111)$$

while the coupled social state approaches $q^* \approx 0.363$.

At $t = 0$ the exact information quantities are

$$I(S_0; Y_0) \approx 0.1265 \text{ nats}, \quad (112)$$

$$I(S_0; U_0) \approx 0.0589 \text{ nats}, \quad (113)$$

$$I(U_0; X_1 | S_0) \approx 0.0341 \text{ nats}. \quad (114)$$

The Pinsker numerator in Eq. (80) is approximately 0.0286 nats, giving $\eta_{\text{IR}}(0) \approx 0.836$. By round 30, the corresponding values are approximately 0.1792, 0.0844, 0.0302, and 0.0232 nats, respectively, with $\eta_{\text{IR}} \approx 0.769$. Figure 13 shows the exact round dependence.

Finally, Fig. 14 varies only the epistemic resistance g and recomputes the exact stationary distribution. The result is the qualitative behavior desired from the model: a high-information population becomes progressively harder for a false controller to steer.

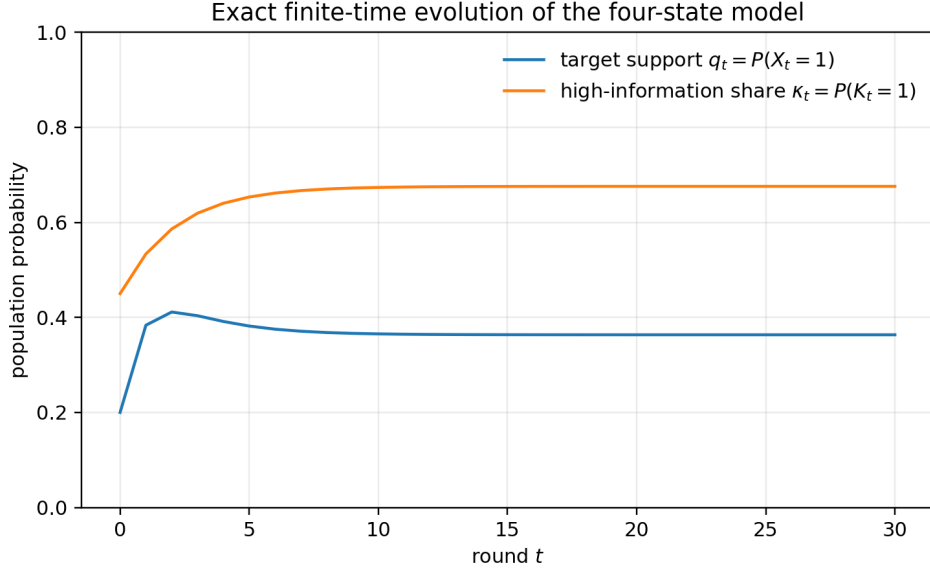


Figure 12: Exact finite-time population dynamics obtained from $\mu_{t+1} = \mu_t M$. No Monte Carlo sampling is used.

11 What has and has not been established

The four-state construction already provides several useful exact statements:

1. The sensing information is exactly

$$I(S_t; Y_t) = h(\epsilon + (1 - 2\epsilon)q_t) - h(\epsilon).$$

2. The controller cannot encode more state information in its actions than it obtained from sensing:

$$I(S_t; U_t) \leq I(S_t; Y_t).$$

3. The epistemically conditioned action-to-population information is available in closed form:

$$I(U_t; X_{t+1} \mid X_t, K_t) = \sum_{x,k} \mu_t(x, k) \{h(\bar{p}_{xk}) - (1 - a_x)h(p_{xk0}) - a_x h(p_{xk1})\}.$$

4. Omitting K_t necessarily loses or preserves, but cannot add, action-to-next-state information under this model:

$$I(U; X' \mid X, K) \geq I(U; X' \mid X).$$

5. Susceptibility and CMI are linked by a rigorous bounded efficiency:

$$2 \sum_s \mu_t(s) a_s (1 - a_s) \chi_s^2 \leq I(U_t; X_{t+1} \mid S_t).$$

6. The behavioral advantage of feedback over a matched open-loop policy is bounded by sensing information:

$$2\Delta_{\text{fb}}^2 \leq I(S_t; U_t) \leq I(S_t; Y_t).$$

What we have *not* yet established is a stochastic-thermodynamic entropy balance of the form “system entropy change + currents + information = total entropy production.” Such a statement requires

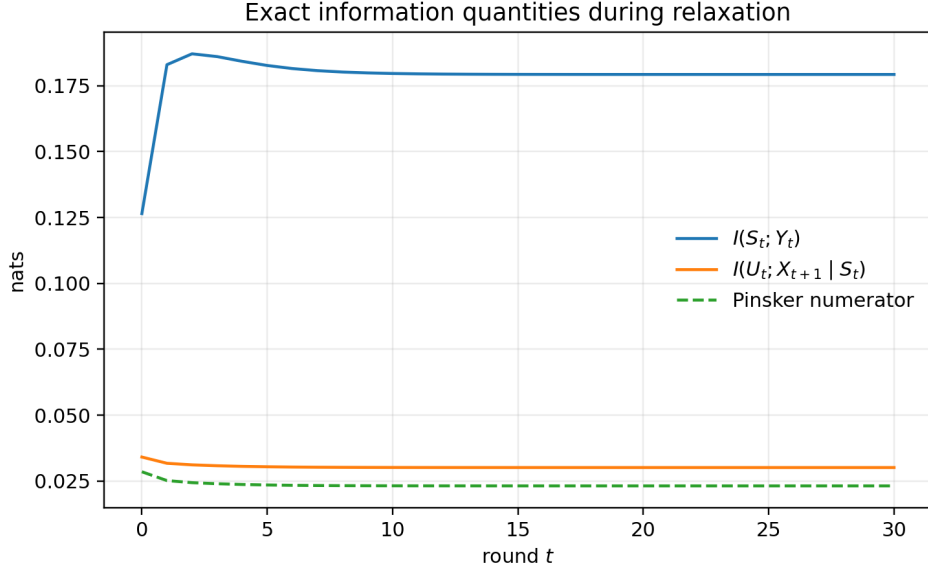


Figure 13: Exact sensing information, action-to-population CMI, and the Pinsker response numerator during relaxation of the reference model. The bound remains satisfied pointwise in time.

an explicit definition of reverse trajectories and, if one wants a thermodynamic interpretation rather than only a KL identity, a principled local-detailed-balance-like structure. This caution is consistent with the distinction emphasized in feedback thermodynamics between different information measures and their operational measurement implementations [1]. The social-imitation framework of Irisarri *et al.* provides a possible route for the later stage: define microscopically reversible social and epistemic transitions first, then derive path irreversibility and second-law-like inequalities from those transition ratios rather than postulating an energy function [2].

12 How this becomes the Stage-II mean-field theory

The next theory should replace the binary epistemic variable K by an epidemic-like distribution of explicit facts. Conceptually, the four-state objects already identify what must survive the generalization:

- X_t becomes a population vote fraction or magnetization m_t ;
- K_t becomes one or more coarse epistemic coordinates κ_t derived from fact occupancy;
- α and δ become fact-acquisition and forgetting terms generated by dashboard sampling and persistence;
- the exact finite-state kernel becomes a mean-field map plus finite- N noise;
- $I(S; Y)$, $I(S; U)$, $I(U; X' | S)$, χ , and the two bounds derived above remain the quantities to reproduce or approximate.

The benefit of the present dummy theory is that the meaning of each information quantity is fixed before the state space becomes large.

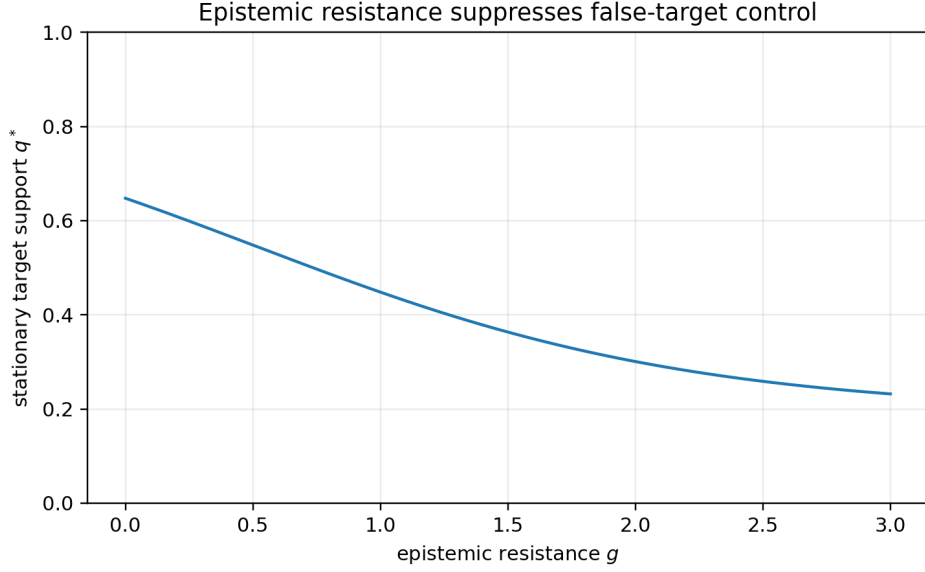


Figure 14: Exact stationary target support versus epistemic resistance. Increasing g shifts the controlled population away from the false controller target while leaving the autonomous K -process stationary fraction unchanged.

13 Summary

The four-state model is intentionally minimal, but it already separates three logically distinct stages of feedback control:

$$\begin{array}{lcl}
 \text{sensed information: } I(S; Y) & \longrightarrow & \text{state-dependent action: } I(S; U) \\
 & \longrightarrow & \text{population response: } I(U; X' | S), \chi
 \end{array} \tag{115}$$

The first data-processing inequality limits how much sensed information can survive in the policy. The second family of inequalities limits how much of the action-dependent information can appear as a coherent population shift. These results provide a clean exact reference for later mean-field and stochastic-thermodynamic extensions with explicit epistemic fact dynamics.

References

- [1] J. M. Horowitz and H. Sandberg, “Second-law-like inequalities with information and their interpretations,” arXiv:1409.5351 (2014).
- [2] L. Irisarri, L. Trigo, R. Toral, and G. Manzano, “Stochastic Thermodynamics of Social Imitation beyond Energetics,” arXiv:2511.14006v2 (2026).